Chapter-4 :- Database Programming With ADO.NET

4	Database Programming with ADO.NET	• Concept of Connected and Disconnected Architecture • Data Providers in ADO.NET • Connection Object • Connected Architecture • Command • Data Reader • Disconnected Architecture • Data Adapter • DataSet • DataTable • DataRow • DataColumn • DataRelation • DataView • Data Binding • Gridview Programming

(1) ADO .NET :

🔷 What is ADO.NET?

ADO.NET is a technology in .NET used to work with databases.

It helps us to:

- Read data from database
- Store data
- Update data

👉 It gives many ready-made classes (tools) to work with data easily.

🔷 Database Support

ADO.NET supports many types of databases:

- It has special classes for SQL Server database
- It also supports OLE DB data sources
- And we can also connect other databases using ADO.NET

👉 So, it is flexible and works with almost any database.

🔷 What ADO.NET Can Do?

ADO.NET is a big collection of classes in .NET.
Using ADO.NET, we can:

- Retrieve (get) data from database
- Manipulate (change/edit) data
- Update data back to the database

👉 And we can do all this in many different ways.

✅ Advantages of ADO.NET

🔶 1. Works with Any Database

You can use ADO.NET to connect with:

- SQL Server
- Oracle
- Access
- Any other database

👉 So, it is very flexible.

🔶 2. Supports Connected and Disconnected Architecture

ADO.NET supports two ways of working:

✓ Connected Architecture

- Connection stays open with database
- Data is used directly

✓ Disconnected Architecture (NEW FEATURE)

- Data is copied and used without continuous connection
- Connection is closed after getting data

👉 This feature is new and very useful in ADO.NET.

🔶 3. Supports XML

ADO.NET supports XML (Extensible Markup Language)

- XML is widely used for data transformation
- It helps in faster data transfer and processing

👉 Because of XML, data handling becomes faster and easier.

🔶 4. Fast, Scalable and Secure

Using ADO.NET, we can build applications that are:

- Fast → Works quickly
- Scalable → Can handle large number of users/data
- Secure → Keeps data safe

🔶 5. Supports Multiple Languages

ADO.NET supports multiple programming languages

👉 Because .NET framework itself supports many languages like:

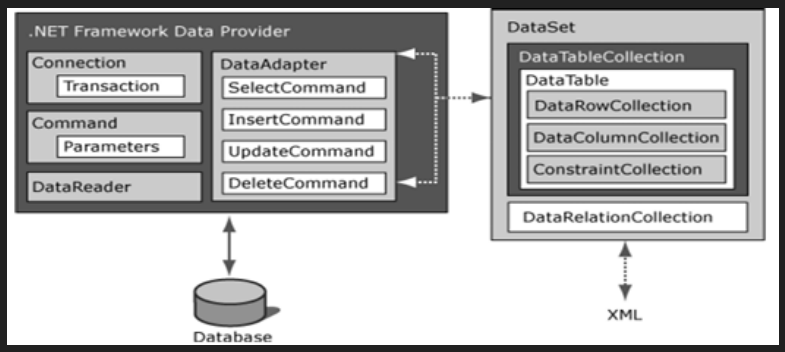
- C#
- VB.NET
- etc.

So, ADO.NET also works with different languages.

✅ Final Easy Definition (Write in Exam)

👉 ADO.NET is a part of .NET framework that provides a set of classes to connect with databases, retrieve data, manipulate data, and update data sources. It supports multiple databases, connected and disconnected architecture, XML data handling, and helps to build fast, scalable, and secure applications with multi-language support.

2) Architecture of ADO .NET:



✅ Architecture of ADO.NET (Simple + Brain Friendly)

ADO.NET has two types of architecture:

1. Connected Architecture

2. Disconnected Architecture

🔷 1) Connected Architecture

✓ Simple Meaning

Connected Architecture means:
👉 You are continuously connected to the database while doing all operations.

✓ Explanation

- In this method, the connection with database stays open all the time
- You directly work with database while connection is active

👉 So, data is fetched row by row directly from database

✓ Advantage

- It is faster than disconnected architecture

❌ Problems

- It creates more traffic on database server
  👉 Because many users stay connected at the same time
- It blocks database memory
  👉 Some part of database memory is used continuously
- Performance becomes slower when many users are connected
- Client memory is not used (everything depends on database)

✓ Important Point

👉 DataReader is used in Connected Architecture

- It keeps connection open
- It reads data one row at a time

✓ Classes Used in Connected Architecture

- Connection
  👉 Used to connect with database
- Command
  👉 Used to execute SQL commands like:
  - Insert
  - Update
  - Delete
  - Select
- DataReader
  👉 Used to read data directly from database (row by row)

🔷 2) Disconnected Architecture (Definition + Usage)

🔷 Definition

👉 Disconnected Architecture is a method in ADO.NET where:

- Data is first taken from the database
- Then stored in local memory
- After that, connection is closed
- You can work on data without staying connected

👉 Later, data is updated back to the database when needed.

✅ What is DataSet? (Definition + Usage)

🔷 Definition

👉 DataSet is a memory-based data storage in ADO.NET.
👉 It stores data in the form of tables inside memory (RAM).

🔷 Usage (What it does)

- Stores all data at once (not row by row)
- No need to keep database connection open

- Can store:
  - One table
  - Or multiple tables

✓ Advantages

- Less load on database (less traffic)
- Better performance for many users
- Uses client memory (RAM) instead of database

✅ Classes Used in Disconnected Architecture

(Definition + Usage)

🔶 1) Connection

✓ Definition

👉 Connection is used to connect application with database

✓ Usage

- Open connection to database
- Close connection after getting data

🔶 2) DataAdapter

✓ Definition

👉 DataAdapter is a bridge (mediator) between database and DataSet

✓ Usage

- Fetches data from database
- Fills data into DataSet / DataTable
- Sends updated data back to database

🔶 3) DataSet

✓ Definition

👉 DataSet is a collection of tables stored in memory

✓ Usage

- Stores data temporarily in memory
- Can handle multiple tables
- Used for working offline (without connection)

🔶 4) DataTable

✓ Definition

👉 DataTable represents one table in memory

✓ Usage

- Stores data of only one table
- Used when you don't need multiple tables

🔶 5) DataRow

✓ Definition

👉 DataRow represents one single row of a table

✓ Usage

- Add new row
- Delete row
- Update (modify) row

🔶 6) DataColumn

✓ Definition

👉 DataColumn represents one column of a table

✓ Usage

- Add new column
- Modify existing column

✅ Final Exam Answer (Short Paragraph)

👉 Disconnected Architecture in ADO.NET allows data to be retrieved from the database and stored in memory using DataSet. After retrieving data, the connection is closed and operations like Create, Read, Update, and Delete can be performed in memory. DataAdapter acts as a bridge between database and DataSet. DataSet stores multiple tables, DataTable stores a single table, DataRow represents rows, and DataColumn represents columns. This architecture reduces database load and improves performance.

⚡ Advantages (Pros)

- ✓ Can work offline
- ✓ Less load on database server
- ✓ Multiple users supported
- ✓ DataSet supports multiple tables
- ✓ Good for Windows Forms

❌ Disadvantages (Cons)

- ❌ Uses more RAM (stores full data)
- ❌ Data may become old (stale data)

💡 Example:
👉 Someone else updates DB → your data still old

👉 Final Easy Summary (Write in Exam)

👉 ADO.NET has two architectures: Connected and Disconnected. In Connected Architecture, the connection with database remains open and data is read directly using DataReader. It is fast but creates more load on database. In Disconnected Architecture, data is stored in memory using DataSet, and connection is closed. It reduces database load and allows CRUD operations in memory. Classes like Connection, Command, DataReader are used in Connected Architecture, while

Connection, DataAdapter, DataSet, DataTable, DataRow, and DataColumn are used in Disconnected Architecture.

3) ✅ Difference Between Connected and Disconnected Architecture:

◈ Connected Architecture	◈ Disconnected Architecture
👉 In this, you stay connected to database all the time during operation	👉 In this, you take a copy of data and work in local memory
👉 Data is always stored and used directly in database	👉 Data is stored in local memory (RAM) while working
👉 Creates heavy traffic on database server when many users are connected	👉 Does not create heavy traffic because users work on local copy
👉 Uses database server memory	👉 Uses client (user) memory
👉 Used for quick operations like: Insert, Update, Delete	👉 Used when you want data mainly for display or reading purpose
👉 Uses Command and DataReader objects	👉 Uses DataAdapter, DataSet, DataTable, DataRow, DataColumn
👉 DataReader is used to fetch data (row by row)	👉 DataSet / DataTable is used to fetch all data at once

✅ Easy Trick to Remember

👉 Connected = Live connection (Direct work on database)
👉 Disconnected = Copy data (Work in memory)

✅ Final Exam Answer (Short Paragraph)

👉 Connected Architecture keeps the database connection open during operations and works directly on database, which is faster but creates heavy load on server. Disconnected Architecture retrieves data into local memory using DataSet and closes the connection, allowing operations without continuous connection. It reduces database traffic and improves performance. Connected Architecture uses Command and DataReader, while Disconnected Architecture uses DataAdapter, DataSet, DataTable, DataRow, and DataColumn.

4) Data Providers in ADO.NET (Simple + Exam Ready)

✅ What are Data Providers?

🔷 Definition

👉 Data Providers are a set of classes in ADO.NET
👉 They are used to connect and work with different databases

🔷 What They Do (Usage)

Data Providers help us to:

- Connect to database
- Execute commands (Insert, Update, Delete, Select)
- Read data
- Send data back to database

👉 They act like a bridge between application and database

✅ Types of Data Providers in ADO.NET

There are 4 main types 👇

🔶 1) SQL Server Provider

✓ Definition

👉 Special provider used to connect only with SQL Server database

✓ Usage

- Works with SQL Server (version 7.0 or later)
- Direct connection with SQL Server

✓ Advantage

- Very fast because it directly works with SQL Server

🔶 2) Oracle Provider

✓ Definition

👉 Special provider used to connect only with Oracle database

✓ Usage

- Works with Oracle (version 8i or later)
- Direct connection with Oracle

✓ Advantage

- Very fast because it directly works with Oracle

🔶 3) OLE DB Provider

✓ Definition

👉 Provider used to connect any database that supports OLE DB driver

✓ Usage

- Can connect to many databases
- Uses OLE DB driver

✓ Advantage

- Flexible → can work with different databases

🔶 4) ODBC Provider

✓ Definition

👉 Provider used to connect any database that supports ODBC driver

✓ Usage

- Works with databases having ODBC (Open Database Connectivity) driver

✓ Advantage

- Can connect to many types of databases

✅ Easy Trick to Remember

👉 SQL + Oracle = Specific (only one database)
👉 OLEDB + ODBC = General (many databases)

✅ Final Exam Answer (Short Paragraph)

👉 Data Providers in ADO.NET are a set of classes used to connect and interact with different databases like SQL Server, Oracle, Access, etc. There are four types of providers: SQL Server Provider and Oracle Provider, which are used for specific databases and provide faster performance, and OLEDB and ODBC Providers, which are used to connect multiple types of databases using drivers. These providers act as a bridge between the application and the database.

*)WHAT IS DATASET :- In simple terms, a DataSet in C# is like a mini, portable database that lives inside your computer's memory while your program is running.

5) Connection in ADO.NET (Simple + Brain Friendly)

🔷 What is Connection? (Definition)

👉 A Connection is used to connect your application with a database

👉 It is the first and most important step in any database program

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🔷 Explanation

- Before working with database, we must open a connection
- ADO.NET provides special classes for this:
  - SqlConnection → for SQL Server
  - OleDbConnection → for OLE DB databases

✓ SqlConnection

👉 Part of SQL Server Data Provider
👉 Used only for SQL Server (7.0 or later)
👉 Gives better performance (fast connection)

✓ OleDbConnection

👉 Part of OLE DB Data Provider
👉 Used to connect any database that supports OLE DB

✅ Properties of Connection Object

👉 Properties are used to get or set values

Property	Description (Simple)
CommandTimeout	Time (in seconds) to wait for a command to execute
ConnectionString	Contains details to connect database (server, user, password, etc.)
ConnectionTimeout	Time (in seconds) to wait while opening connection
DefaultDatabase	Default database name
Mode	Access permission (read/write)
Provider	Name of database provider
State	Shows connection status (Open or Closed)
Version	Shows ADO version number

✅ Methods of Connection Object

👉 Methods are used to perform actions

Method	Description (Simple)
BeginTrans	Starts a new transaction
Close	Closes the connection
CommitTrans	Saves changes and ends transaction
Open	Opens the connection
RollbackTrans	Cancels changes and ends transaction

✅ Events of Connection Object

👉 Events are actions that happen automatically

Event	Description (Simple)
BeginTransComplete	Runs after transaction starts
CommitTransComplete	Runs after transaction is saved
ConnectComplete	Runs after connection starts
Disconnect	Runs after connection ends
ExecuteComplete	Runs after command execution
RollbackTransComplete	Runs after canceling transaction
WillConnect	Runs before connection starts
WillExecute	Runs before command execution

✅ Easy Understanding

👉 Connection = Bridge between application and database

👉 Steps:

1. Open connection
2. Do database work
3. Close connection

✅ Final Exam Answer (Short Paragraph)

👉 Connection in ADO.NET is used to establish a link between an application and a database. It is the first step in database programming. ADO.NET provides classes like SqlConnection for SQL Server and OleDbConnection for OLE DB data sources. The connection object has properties like ConnectionString and State, methods like Open, Close, BeginTrans, CommitTrans, and RollbackTrans, and events like ConnectComplete and ExecuteComplete. It helps in managing database communication efficiently.

6) Command in ADO.NET (Simple + Brain Friendly)

🔷 What is Command? (Definition)

👉 **Command** object is used to **execute SQL queries on a database**

👉 It is the **main (heart) class** in Data Provider

🔷 **Usage (What it does)**

Using Command object, we can run SQL queries like:

- INSERT → add data
- UPDATE → change data
- DELETE → remove data
- SELECT → get data

👉 It is mostly used in **Connected Architecture**

🔷 **Important Rule**

👉 Before using Command:

- You must **open connection** using Open()

👉 After work is done:

- You must **close connection** using Close()

❌ Command does NOT open/close connection automatically

✅ **Properties of Command Object**

Property	Description (Simple)
CommandText	SQL query (like SELECT, INSERT, etc.) *OR* Contains the text of a SQL query
CommandTimeout	Time (in seconds) to wait for query
CommandType	Specifies the type of command to be executed
Connection	Connection object used to connect database OR Specifies the connection to the database
Parameters	Used to pass values in query *OR* Specifies a collection of parameters for the SQL query

Transaction	Used to execute query inside transaction

✅ Methods of Command Object

👉 Methods are actions performed by Command

🔶 **1) Cancel()**

👉 Stops the running query

🔶 **2) CreateParameter()**

👉 Creates a parameter for SQL query
👉 Used to pass values safely

🔶 **3) ExecuteNonQuery()**

✓ **Usage**

👉 Used for:

- INSERT
- UPDATE
- DELETE

✓ **Output**

- Returns **integer value**

👉 If value > 0 → Success
👉 If value = 0 → No effect

🔶 **4) ExecuteReader()**

✓ **Usage**

👉 Used for **SELECT query**

👉 Returns data in **DataReader**

👉 Reads data **row by row**

🔶 **5) ExecuteScalar()**

✓ **Usage**

👉 Used for SELECT query

👉 Returns **only one value**

👉 That value is:

- First row
- First column

✓ **Example**

If query is:

SELECT * FROM student

👉 It will return only **first student's first column value**

✅ Event of Command Object :

Event	Description (Simple)
StatementComplete	Runs after SQL command is fully executed

✅ **Easy Understanding**

👉 Command = **Tool to run SQL queries**

👉 Types of output:

- No data → ExecuteNonQuery
- Full data → ExecuteReader
- Single value → ExecuteScalar

✅ **Final Exam Answer (Short Paragraph)**

👉 Command object in ADO.NET is used to execute SQL queries like Insert, Update, Delete, and Select. It is mainly used in Connected Architecture. Before executing any command, the connection must be opened using Open() and closed after execution. It has properties like CommandText and Connection, and methods like ExecuteNonQuery, ExecuteReader, and ExecuteScalar. It also has events like StatementComplete which occurs after execution of a query.

7) **DataReader in ADO.NET (Simple + Brain Friendly)**

🔷 **What is DataReader? (Definition)**

👉 **DataReader** is a lightweight and fast object used to **read data from a database**

👉 It is **powerful but simple**, mainly used for reading data

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🔷 **Usage (What it does)**

- Used in **Connected Architecture**
- Works together with **Command object**
- Used to **fetch data from database**

🔷 **Important Points**

👉 Reads data in **sequential (forward-only) manner**

- You can go **only forward (next row)**
- ❌ You cannot go backward
- ❌ You cannot jump to a specific row

👉 Data is read **one row at a time**

✅ **Properties of DataReader Object**

👉 Properties are used to get information

Property	Description (Simple)

FieldCount	Number of columns (fields) in result
IsClosed	Shows whether DataReader is closed (True/False)
Item	Gets value of a column (by name or index number)
RecordsAffected	Number of rows affected by query
HasRows	Checks if data exists (True if rows are present)

✅ **Methods of DataReader Object**

👉 Methods are actions

🔶 **1) Close()**

👉 Closes the DataReader
👉 Must be called after reading data

🔶 **2) IsDBNull()**

👉 Checks if a field value is **NULL (empty)**
👉 Returns True if value is NULL

🔶 **3) Read()**

👉 Moves to **next row** and loads data

✓ **Important**

- Each call reads **one row**
- Used in loop to read all rows

✓ **Accessing Data**

👉 After using Read(), you can get column values:

- By **column name**
- By **index number (position)**

✅ **Easy Understanding**

👉 DataReader = **Fast data reader (one row at a time)**

👉 Think like:

- Open connection
- Run SELECT query
- Read row by row
- Close reader

✅ **Final Exam Answer (Short Paragraph)**

👉 DataReader in ADO.NET is a lightweight object used to read data from a database in Connected Architecture. It works with Command object and reads data in forward-only and sequential manner, meaning it reads one row at a time and cannot move backward. It has properties like FieldCount and HasRows, and methods like Read(), Close(), and IsDBNull(). It is fast and efficient for reading data.

8) **DataAdapter in ADO.NET (Simple + Brain Friendly)**

🔷 **What is DataAdapter? (Definition)**

👉 **DataAdapter** is used in **Disconnected Architecture**

👉 It is a very important object because:

- It helps to **get data from database**
- It also helps to **send updated data back to database**

🔷 **Explanation (What it does)**

👉 DataAdapter works like a **bridge (middle object)** between:

- Database
- Local memory (DataSet / DataTable)

👉 It allows you to:

- Fetch data from database → store in client memory
- Save updated data → back to database

🔷 **Important Concept**

👉 In Disconnected Architecture:

- Data is stored in **DataSet or DataTable (local memory)**
- Not directly in database

👉 DataAdapter is used to:

- Fill data into DataSet / DataTable
- Update data back to database

✅ **Main Tasks of DataAdapter**

👉 DataAdapter performs **two important tasks**:

🔶 **1) Fill Data**

👉 It **fetches data from database**
👉 Stores it into:

- DataSet
- DataTable

🔶 **2) Update Data**

👉 It sends **modified data back to database**

✅ **Working Process (Step-by-Step)**

1. DataAdapter connects to database

2. Fetches required data
3. Stores data into **DataSet / DataTable (local memory)**
4. Connection is **automatically closed**

👉 Now:

- Data is available in local memory
- No need to stay connected

🔷 **Local Operations**

👉 After data is loaded, we can perform operations like:

- INSERT
- UPDATE
- DELETE
- SELECT

👉 All operations are done **locally in DataSet or DataTable**

✅ **Important Point**

👉 After filling data:

- Connection is **automatically closed**
- But data remains available in memory

✅ **Easy Understanding**

👉 DataAdapter = **Bridge between Database and Memory**

👉 Flow:
Database ⇄ DataAdapter ⇄ DataSet/DataTable

✅ **Final Exam Answer (Short Paragraph)**

👉 DataAdapter in ADO.NET is used in Disconnected Architecture and acts as a bridge between database and local memory. It is used to fetch data from database into DataSet or DataTable and also to update modified data back to database. It performs two main tasks: filling data and updating data. After filling data, the connection is automatically closed, and all operations like Insert, Update, Delete, and Select are performed locally in DataSet or DataTable.

✅ **Properties of DataAdapter Object (Simple + Full)**

👉 Properties are used to **automatically manage SQL commands in background**

🔶 **1) SelectCommand**

✓ **Explanation**

👉 This property works in **background**
👉 It automatically creates the **SELECT command**

👉 **OleDbCommandBuilder** class works behind it
👉 It generates SELECT query automatically

🔶 **2) DeleteCommand**

✓ **Explanation**

👉 This also works in **background**
👉 It creates **DELETE command automatically**

👉 If you delete rows from:

- DataSet
- DataTable

👉 Then OleDbCommandBuilder:

- Generates DELETE query
- Stores it in this property

🔶 **3) InsertCommand**

✓ **Explanation**

👉 This works in **background**
👉 It creates **INSERT command automatically**

👉 If you add new rows in:

- DataSet
- DataTable

👉 Then OleDbCommandBuilder:

- Generates INSERT query
- Stores it here

🔶 **4) UpdateCommand**

✓ **Explanation**

👉 This works in **background**
👉 It creates **UPDATE command automatically**

👉 If you modify data in:

- DataSet
- DataTable

👉 Then OleDbCommandBuilder:

- Generates UPDATE query
- Stores it here

✅ **Methods of DataAdapter Object (Very Important)**

🔶 **1) Fill() Method**

✓ **Definition**

👉 Most important method of DataAdapter

✓ **Usage**

👉 Used to **load data from database into:**

- DataSet

- DataTable

✓ **Important Points**

- Only **SELECT command** is used here
- It fills data into memory

✓ **Extra Explanation**

👉 Fill method:

- Adds data from database → into DataSet

👉 It can take parameters like:

- DataSet object
- Table name (alias)
- Starting record number
- Ending record number

🔶 **2) Update() Method**

✓ **Definition**

👉 Most important method

✓ **Usage**

👉 Used to **send updated data back to database**

✓ **Important Points**

👉 Before using Update():

- You must use **SqlCommandBuilder**

👉 Why?

- It automatically creates:
    - INSERT

- UPDATE
- DELETE
- ALTER commands

✓ **Working**

👉 Update method checks each row in DataSet:

- New row → INSERT
- Modified row → UPDATE
- Deleted row → DELETE

✓ **Ways to Call Update()**

You can pass:

- Array of DataRow
- DataSet
- DataSet + Table name

✅ **Events of DataAdapter (Very Simple Understanding)**

👉 First understand:

👉 **Event = something that happens automatically**

🔶 **1) FillError (Very Simple)**

✓ **Meaning**

👉 When DataAdapter is **loading data (Fill)**

👉 If any **error happens** → this event runs

🔶 **2) RowUpdating (Before Update)**

✓ **Meaning**

👉 When you use Update() method

👉 Before each row is updated in database → this event runs

✓ **Simple Flow**

- Row is going to update ❗
- BEFORE updating → **RowUpdating event**

🔶 **3) RowUpdated (After Update)**

✓ **Meaning**

👉 After each row is updated → this event runs

✓ **Simple Flow**

- Row is updated ✅
- AFTER updating → **RowUpdated event**

✓ **Example**

👉 You changed 5 rows

👉 Then:

- Row 1 updated → RowUpdated
- Row 2 updated → RowUpdated

👉 (after each update)

✅ **Super Easy Trick (Remember in Exam)**

👉 Think like this:

- ❌ Error while filling → **FillError**
- ⏳ Before update → **RowUpdating**

- ✅ After update → **RowUpdated**

✅ **Easy Understanding (MOST IMPORTANT)**

👉 DataAdapter does 2 main things:

🔷 **1) Fill()**

👉 Used for **SELECT**

👉 Means:

- Take data from database
- Put into DataSet / DataTable

🔷 **2) Update()**

👉 Used for:

- INSERT
- UPDATE
- DELETE

👉 Means:

- Send changed data → back to database

🔷 **CommandBuilder (Very Simple)**

👉 You don't write SQL manually

👉 CommandBuilder automatically creates:

- INSERT
- UPDATE
- DELETE

👉 So your work becomes easy

✅ **Final Simple Line (Exam Ready)**

👉 DataAdapter events include FillError, which occurs when an error happens during filling data, RowUpdating which occurs before updating each row, and RowUpdated which occurs after updating each row. DataAdapter uses Fill() method for retrieving data and Update() method for sending changes back to database, while CommandBuilder automatically generates SQL commands.

9) ✅ DataSet in ADO.NET (Simple + Brain Friendly)

🔷 What is DataSet? (Definition)

👉 **DataSet** is an object that stores **data in memory (RAM)**

👉 It represents a **group of database tables in memory**

🔷 Explanation

- DataSet can contain:
  - One table
  - Or multiple tables

👉 So, it is a **collection of DataTables**

👉 DataSet is the **main part of Disconnected Architecture**

- It stores complete data:
  - Tables
  - Relationships
  - Constraints

👉 Important point:

❌ DataSet does NOT keep connection with database
👉 It works **without connection (disconnected)**

🔷 What You Can Do with DataSet

👉 You can:

- Access data
- Modify data
- Update data
- Delete data

👉 After changes:

- Data can be **sent back to database**

🔷 Advantage

👉 Because DataSet is disconnected:

- Less load on database
- Less use of database connection
- Less record locking
- Better performance

✅ Properties of DataSet Object

👉 Properties give information about DataSet

Property	Description (Simple)
DataSetName	Gets or sets the name of DataSet
HasErrors	Checks if any table has errors
IsInitialized	Checks if DataSet is ready to use
Relations	Shows relationships between tables

Tables	Gets the collection of tables contained/Ava in the DataSet.

✅ **Methods of DataSet Object**

👉 Methods are actions

🔶 **1) AcceptChanges()**

👉 Saves all changes made in DataSet
👉 Marks changes as final

🔶 **2) Clear()**

👉 Removes all data from DataSet
👉 Deletes all DataTables data

🔶 **3) ReadXml()**

👉 Reads data and structure from XML file
👉 Loads into DataSet

🔶 **4) ReadXmlSchema()**

👉 Reads only **structure (schema)** from XML

🔶 **5) RejectChanges()**

👉 Cancels all changes
👉 Restores original data

🔶 **6) Select()**

👉 Used to **filter or select specific data**

👉 It is an **overloaded method**
👉 Means:

- Can be used in different ways

🔶 **7) WriteXml()**

👉 Saves DataSet data into XML file

🔶 **8) WriteXmlSchema()**

👉 Saves only structure (schema) into XML

✅ **Event of DataSet Object**

Event	Description (Simple)
Initialized	Occurs after DataSet is ready to use

✅ **Easy Understanding**

👉 DataSet = **Mini database in memory**

👉 You:

- Load data → work offline
- Make changes → send back later

✅ **Final Exam Answer (Short Paragraph)**

👉 DataSet in ADO.NET is an in-memory representation of data that can contain multiple tables. It is mainly used in disconnected architecture and does not maintain a connection with the database. It allows operations like insert, update, delete, and select on data stored in memory. It has properties like Tables and Relations, methods like AcceptChanges, RejectChanges, ReadXml, and WriteXml, and event Initialized. It improves performance by reducing database load and connection usage.

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